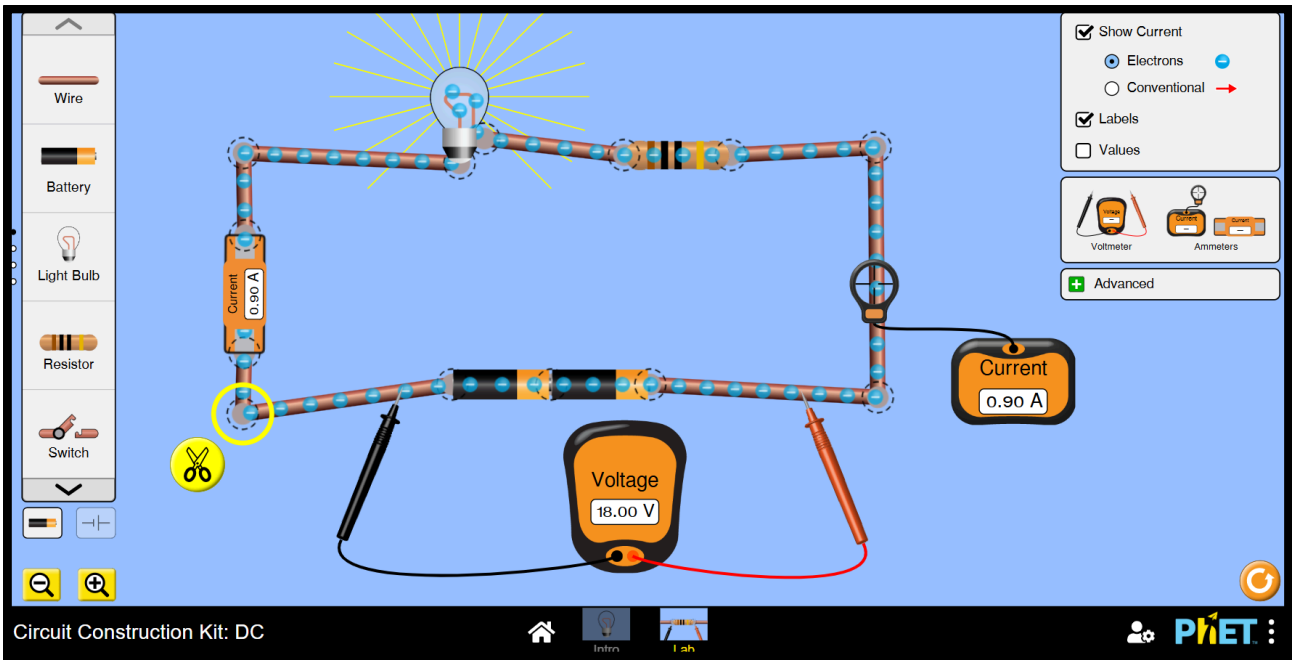
PhET Simulation Worksheet for Virtual Lab: Circuit Construction Kit (DC)



Name: _____

Date: _____

Class: _____

PhET Simulation Worksheet for the Virtual Lab:

Circuit Construction Kit (DC)

Name: _____

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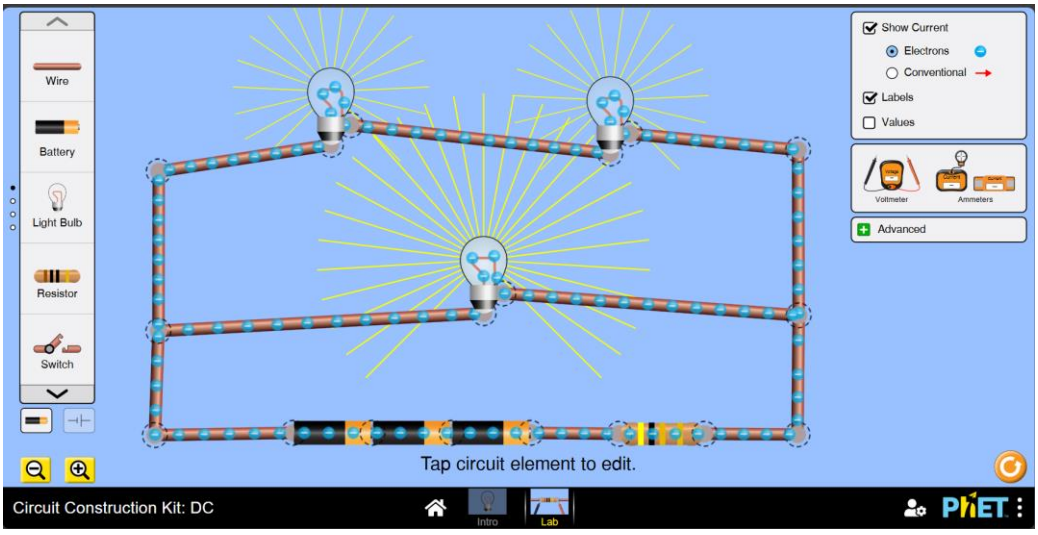
Class: _____

Objective:

Explore basic electrical concepts and circuit designs using the PhET Circuit Construction Kit (DC) simulation.

Instructions:

1. Go to the PhET Simulation Circuit Construction Kit (DC) website:
   https://phet.colorado.edu/en/simulations/circuit-construction-kit-dc
2. Follow the steps below to construct and analyze different circuits.
3. Record your observations and answers.



Part 1: Basic Components and Measurements

1. Construct a Simple Circuit

- Build a circuit with a battery, a light bulb, and wires to connect them.
- Set the battery voltage to 9V.

Questions:

1.1 Describe what happens to the light bulb when the circuit is complete.

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1.2 Measure the current using the ammeter tool in the simulation. What is the current?

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Part 2: Ohm's Law

2. Ohm's Law with Resistors

- Replace the light bulb with a resistor

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- Set the resistor value to 10 ohms.

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Questions:

2.1 What is the current flowing through the resistor? (Use the ammeter tool)

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2.2 Calculate the current using Ohm's Law (I = V/R). Show your work.

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Part 3: Series Circuits

3. Construct a Series of Circuit

- Build a circuit with a battery and two resistors in series.
- Set the battery voltage to 9V.
- Set each resistor to 10 ohms.

Questions:

3.1. What is the total resistance in the circuit?

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3.2. Measure the current flowing through the circuit. What is the current?

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3.3. Calculate the current using Ohm's Law. Show your work.

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Part 4: Parallel Circuits

4. **Construct a Parallel Circuit**
   - Build a circuit with a battery and two resistors in parallel.
   - Set the battery voltage to 9V.
   - Set each resistor to 10 ohms.

Questions:

4.1. What is the equivalent resistance of the parallel circuit?

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4.2. Measure the total current supplied by the battery. What is the current?

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4.3. Calculate the total current using Ohm's Law. Show your work.

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Part 5: Combination Circuits

5. Construct a Combination Circuit

- Build a circuit with a battery, two resistors in series, and a third resistor in parallel with one of the series resistors.
- Set the battery voltage to 9V.
- Set each resistor to 10 ohms.

Questions:

5.1. What is the total resistance of the combination circuit?

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5.2. Measure the total current supplied by the battery. What is the current?

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5.3. Calculate the total current using Ohm's Law. Show your work.

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Answer Key

Part 1: Basic Components and Measurements

1.1. Observation: **The light bulb lights up when the circuit is complete**.

1.2. Current: **The current can be observed using the ammeter tool. The exact value depends on the specifics of the simulation setup, but typically, with a light bulb of resistance around 10 ohms and a 9V battery, the current might be around 0.9 A.**

Part 2: Ohm's Law

2.1. Current through the resistor: **0.9 A**

2.2. Calculation using Ohm's Law:

$$I = \frac{V}{R} = \frac{9V}{10\Omega} = 0.9A$$

Part 3: Series Circuits

3.1. Total resistance in the circuit: **20 ohms**

3.2. Current flowing through the circuit: **0.45 A**

3.3. Calculation using Ohm's Law:

$$I = \frac{V}{R_{total}} = \frac{9V}{20\Omega} = 0.45A$$

Part 4: Parallel Circuits

4.1. Equivalent resistance of the parallel circuit: **5 ohms**

4.2. Total current supplied by the battery: **1.8 A**

4.3. Calculation using Ohm's Law:

$$I = \frac{V}{R_{total}} = \frac{9V}{5\Omega} = 1.8A$$

Part 5: Combination Circuits

5.1. Total resistance of the combination circuit: **Calculation needed based on configuration**.

For two series resistors (20 ohms) and one parallel (10 ohms):

$$R_{eq} = 10\Omega + \left(\frac{1}{\left(\frac{1}{10\Omega} + \frac{1}{10\Omega} \right)} \right) = 15\Omega$$

5.2. Total current supplied by the battery: **0.6 A**

5.3. Calculation using Ohm's Law:

$$I = \frac{V}{R_{total}} = \frac{9V}{15\Omega} = 0.6A$$